Tasfia Akter Sara

+1 (267) 457-8307 | tasfiaakter@cse.mist.ac.bd | tasfiasara1000@gmail.com | Google Scholar | Github
217 N 35th St Philadelphia, PA 19104, USA

Education

- **Military Institute of Science and Technology, Bangladesh University of Professionals** *Dhaka, Bangladesh*
  B.Sc. IN COMPUTER SCIENCE AND ENGINEERING Jan 2020 - Apr 2024
  - CGPA: **3.83/4.00** (ranked 4th out of 88 students)
  - Undergraduate Thesis: An Ensemble-Based Deep Neural Approach for Classifying Skin Allergies
  - IELTS Score: **7.5** (L: 8.5, R: 8.5, W: 7, S: 6)

Research Interest

Human Computer Interaction, Machine Learning, Artificial Intelligence, Digital Image Processing, Data Science, IoT, Computer Vision, Natural Language Processing

Professional Experience

- **Former Lecturer, Department of Computer Science and Engineering** *Dhaka, Bangladesh*
  Military Institute of Science and Technology Apr 2024 – Aug 2026
- **Industrial Trainee** *Dhaka, Bangladesh*
  Brain Station 23 2023
- **Management Lead, MIST Mars Rover Society** *Dhaka, Bangladesh*
  Military Institute of Science and Technology 2022 - 2023

Projects

- **Hydroferma (Capstone Project)** *2022 - 2023*
  Tools and Technologies: Machine Learning, Image Processing, Firebase, Flutter, Python, Arduino, Data Analysis, IoT sensors
  - An automated ML-based hydroponic vertical farm monitoring system (with mobile application)
- **Unlocking Voices** *2023 - 2024*
  Technologies: HCI, HE, AE, WCAG
  - Evaluation of the usability & accessibility of mobile applications for stutter reduction and communication enhancement in children
- **PPDHero** *2023 - 2024*
  Technologies: Information Analysis
  - A research to empower new mothers & raise awareness about postpartum depression
- **NLP Sentiment Robustness Study** *2026*
  Technologies: Natural Language Processing, Hugging Face Transformers, PyTorch, Scikit-learn, TF-IDF, DistilBERT, Python, Google Colab
  - A comparative study of a TF-IDF + Logistic Regression baseline and a fine-tuned DistilBERT model for IMDb sentiment classification, with evaluation of robustness to spelling perturbations, prediction flips, and confidence behavior
- **Unraveling the Threads** *Current*
  Technologies: Machine Learning, Image Processing, Python, Data Analysis
  - A comparative study of jamdani and non-jamdani fabric classification using deep learning
- **The Last of Us** *2023*
  Tools: Blender
  - A graphic-based post-apocalyptic world simulation
- **Conveyor Belt Infrared Counter** *2022*
  Technologies: Arduino and IoT Sensors
  - Conveyor belt infrared counter with Arduino integration for object counting and display using motors, IR sensor, and LED display.
- **MCC Virtual Assistant** *2022*
  Tools and Technologies: HTML, CSS, PHP
  - A web-based management system for MIST Computer Club (MCC).
- **Subscriptia- An E-commerce Application, Student Study Portal** *2022*
  Technology: Python, Django Framework, CSS, Firebase

Technical Skills

- **Programming Languages:** C, C++, Python, Java, PL/SQL, Assembly (Intel ×86)
- **AI/ML & NLP:** Machine Learning, Deep Learning, Digital Image Processing, Computer Vision, Natural Language Processing, Transformer Models, Data Analysis
- **Libraries & Tools:** Hugging Face Transformers, NumPy, Matplotlib, Google Colab, MATLAB, Git, Visual Studio, Code::Blocks, Android Studio
- **Database:** Oracle, Firebase
- **Simulation Softwares:** Cisco Packet Tracer, Circuit Maker
- **Operating Systems:** Linux, Windows
- **Embedded Systems:** Arduino, ESP32
- **Other Tools & Technologies:** Blender, OpenGL, LaTeX, MS Word, MS Project, MS PowerPoint, MS Excel, Canva

Publications

C=Conference, J=Journal on process, S=In Submission, A=Accepted

[C.1] **T. A. Sara**, S. N. Nazifa, S. Tasneem, T. U. Shakib, & M. N. Islam, (2023, December). "PPDHero: Requirements Elicitation and Development of a System to Empower New Mothers on Postpartum Depression." In International Conference on Intelligent Systems Design and Applications (pp. 389-398). Cham: Springer Nature Switzerland

[C.2] S. N. Nazifa, S. T. Mulk, **T. A. Sara**, T. U. Shakib and M. N. Islam, "Unlocking Voices: Assessing the Usability and Accessibility of Mobile Applications for Stutter Reduction," 2024 27th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2024, pp. 1016-1021.

[C.3] **T. A. Sara**, B. Ghosh, S. N. Nazifa and H. Jahan, "An Ensemble-Based Deep Neural Approach for Classifying Skin Allergies," 2025 28th International Conference on Computer and Information Technology (ICCIT), 2025, pp. 3500-3505.

[A.1] T. Mahamud, S. W. Tawassaf, **Sara, T. A**, H. Rahman, and MD. I. Salekin, "Machine Learning Model for Predicting Mechanical Behavior of I-Beams with Varying Dimensions." 3rd ICMMPE 2025, DUET, Dhaka. *(In press)*

[J.1] **T. A. Sara**, K. H. Irfan, M.R.U. Islam and H. Jahan "A Hybrid DINO-ViT and Quantum Annealing Framework for Skin Lesion Classification"

HONORS AND AWARDS

- **Commandant's List of Honour** *2020 - 2024*
  Military Institute of Science and Technology
  - Achieved CGPA 3.80 or above in all the four academic years
- **Dean's List of Honour** *2020 - 2024*
  Military Institute of Science and Technology
  - Achieved CGPA 3.70 or above in all the four academic years
- **Best Technical Paper Presentation** *2024*
  International Conference on Computer and Information Technology (ICCIT)
  - Got the best technical presentation award of the session for the paper titled "Unlocking Voices: Assessing the Usability and Accessibility of Mobile Applications for Stutter Reduction"
- **Champion in University Rover Challenge 2021** *2021*
  The Mars Society, USA
  - Team Member of MIST MONGOL BAROTA
- **2nd Runnerup in Anatolian Rover Challenge 2022** *2022*
  Space Exploration Society(UKET), Turkiye
  - Management Lead of MIST MONGOL BAROTA

Teaching Experience

- **Theory Courses:**
  - CSE - 205 Object Oriented Programming, MIST, Bangladesh, Lecturer(85 students), Spring 2026, Spring 2025
  - CSE - 217 Theory of Computation, MIST, Bangladesh, Lecturer(60 students), Spring 2024
  - CSE - 109 Computer Programming, MIST, Bangladesh, Lecturer(100 students), Fall 2025, Fall 2024
- **Sessional Courses:**
  - CSE - 206 Object Oriented Programming Sessional, MIST, Bangladesh, Lecturer(85 students), Spring 2026, Spring 2025
  - CSE - 204 Data Structure and Algorithm Sessional, MIST, Bangladesh, Lecturer(45 students), Spring 2026
  - CSE - 302 Database Management System, MIST, Bangladesh, Lecturer(45 students), Spring 2024
  - CSE - 308 Operating System, MIST, Bangladesh, Lecturer(45 students), Spring 2024
  - CSE - 106 Structured Programming Language, MIST, Bangladesh, Lecturer(105 students), Fall 2024
  - CSE - 110 Computer Programming, MIST, Bangladesh, Lecturer(100 students), Fall 2025, Fall 2024
  - CSE - 104 Digital Logic Design Sessional (85 students), Fall 2025

Academic References

1. **Dr. Md. Mahbubur Rahman**
   Professor, Computer Science and Engineering, Military Institute of Science and Technology
   Email: mahbub@cse.mist.ac.bd
2. **Dr. Muhammad Nazrul Islam**
   Associate Professor, Computer Science and Engineering, Military Institute of Science and Technology
   Email: nazrul@cse.mist.ac.bd